

Zombie Farm Crop Rebalancing Model

Regular-crop unlock schedule, target profit and experience curves, and fertilized economy analysis. Seasonal crops are intentionally excluded from this working model.

Design intent

Grow time creates an active-versus-passive tradeoff: longer crops earn more per harvest but less per hour. Unlock level raises the absolute reward in discrete integer steps. Current seed prices remain unchanged for this iteration so their effect on fertilization can be evaluated directly.

Core assumptions

All profit figures are net of seed cost and the 10-gold plow cost. Total XP includes one plow XP. Per-hour figures assume immediate harvesting and replanting with no downtime. Rewards use round-half-up to whole game units after level scaling.

Level scaling

Profit adds 65% of the level-1 baseline per 20 levels. Harvest XP adds 38% per 20 levels. Both are linear and non-compounding.

25	regular crops
1-50	proposed level range
2 max	crop unlocks per level

Equations

Time baseline	$t = \text{grow time in hours}$
Base profit	$P0(t) = \max(1, 5.17855 * t^{0.566707} - 1.36057)$
Profit level factor	$MP(L) = 1 + 0.65 * (L - 1) / 20$
Target profit	$P(t, L) = \text{round}(P0(t) * MP(L))$
Base harvest XP	$X0(t) = \max(1, 1.50 * t^{0.30})$
XP level factor	$MX(L) = 1 + 0.38 * (L - 1) / 20$
Total cycle XP	$X(t, L) = \text{round}(X0(t) * MX(L)) + 1 \text{ plow XP}$
Hourly rates	$\text{profit/hour} = P / t; \text{XP/hour} = X / t$
Target harvest value	$V = \text{current seed cost} + 10 + P$
Fertilized profit	$PF = 2V - \text{current seed cost} - 10 = \text{current seed cost} + 10 + 2P$
Fertilized rate	$\text{fertilized profit/hour} = PF / t$

Calibration anchors

The fractional exponents were selected to reproduce the agreed level-1 starter behavior before integer rounding.

Crop	Grow time	Normal profit	Harvest XP	Total cycle XP
Carrots	15m	1	1	2
Tomatoes	4h	10	2	3
Onions	24h	30	4	5

Proposed Unlock Schedule and Normal Economy

The regular crops retain their original relative order. Proposed levels are selected to keep unlocks to one or two crops per level while taking advantage of integer reward breakpoints.

Proposed level	Crop	Current level	Grow time	Current cost	Current value	Current profit	Target profit	Target total XP	Profit/hour	XP/hour
1	Carrots	1	15m	5	16	1	1	2	4	8
1	Onions	1	24h	20	60	30	30	5	1.25	0.21
3	Tomatoes	1	4h	10	30	10	11	3	2.75	0.75
5	Turnips	3	12h	30	56	16	22	4	1.83	0.33
9	Bread Fruit	5	1h	20	33	3	5	3	5	3
10	Potatoes	6	24h	50	100	40	39	6	1.62	0.25
14	Sampaguita	8	30m	25	38	3	3	3	6	6
15	Coffee	9	8h	20	52	22	23	5	2.88	0.62
16	Corn	10	2h	60	76	6	9	3	4.5	1.5
19	Venus Flytrap	11	6h	80	107	17	21	4	3.5	0.67
20	Celery	12	1h	40	54	4	6	3	6	3
21	Spineapple	14	15m	17	29	2	2	2	8	8
23	Broccoli	15	4h	70	92	12	17	4	4.25	1
25	Garlic	16	8h	50	80	20	28	5	3.5	0.62
27	Bloodberry	17	1h	55	70	5	7	3	7	3
29	Cauliflower	18	12h	90	132	32	38	6	3.17	0.5
31	Skellyberry	19	30m	40	56	6	4	3	8	6
33	Lima Beans	20	24h	120	205	75	61	7	2.54	0.29
35	Sun Glower	21	4h	150	180	20	21	5	5.25	1.25
37	Dragon Fruit	22	24h	120	200	70	65	8	2.71	0.33
39	Pumpking	23	8h	142	176	24	35	6	4.38	0.75
41	Corpse Flower	24	6h	200	228	18	30	6	5	1
43	Meat Flower	25	15m	26	39	3	2	3	8	12
45	Eyebiscus	41	12h	100	144	34	48	7	4	0.58
50	Heartchoke	43	24h	125	180	45	78	9	3.25	0.38

Reading the curve: short crops improve from 4 profit/hour at level 1 to 8 profit/hour in the late game, while 24-hour crops remain less efficient but provide much larger single-harvest rewards and increasing total XP.

Fertilized Economy

Fertilization doubles gross harvest value rather than net profit. With current seed prices held fixed, higher initial investment is returned a second time and becomes a major reward lever.

Level	Crop	Grow time	Current cost	Target profit	Target value	Normal profit/hour	Fertilized profit	Fertilized profit/hour
1	Carrots	15m	5	1	16	4	17	68
1	Onions	24h	20	30	60	1.25	90	3.75
3	Tomatoes	4h	10	11	31	2.75	42	10.5
5	Turnips	12h	30	22	62	1.83	84	7
9	Bread Fruit	1h	20	5	35	5	40	40
10	Potatoes	24h	50	39	99	1.62	138	5.75
14	Sampaguita	30m	25	3	38	6	41	82
15	Coffee	8h	20	23	53	2.88	76	9.5
16	Corn	2h	60	9	79	4.5	88	44
19	Venus Flytrap	6h	80	21	111	3.5	132	22
20	Celery	1h	40	6	56	6	62	62
21	Spineapple	15m	17	2	29	8	31	124
23	Broccoli	4h	70	17	97	4.25	114	28.5
25	Garlic	8h	50	28	88	3.5	116	14.5
27	Bloodberry	1h	55	7	72	7	79	79
29	Cauliflower	12h	90	38	138	3.17	176	14.67
31	Skellyberry	30m	40	4	54	8	58	116
33	Lima Beans	24h	120	61	191	2.54	252	10.5
35	Sun Glower	4h	150	21	181	5.25	202	50.5
37	Dragon Fruit	24h	120	65	195	2.71	260	10.83
39	Pumpkin	8h	142	35	187	4.38	222	27.75
41	Corpse Flower	6h	200	30	240	5	270	45
43	Meat Flower	15m	26	2	38	8	40	160
45	Eyebiscus	12h	100	48	158	4	206	17.17
50	Heartchoke	24h	125	78	213	3.25	291	12.12

Largest rates: Meat Flower (160/hour), Spineapple (124/hour), Skellyberry (116/hour), Sampaguita (82/hour), Bloodberry (79/hour)

Interpretation: Short, expensive crops become the strongest fertilized earners; full fertilization creates a separate hierarchy driven heavily by current seed cost. When costs are revisited, they can distinguish safe low-investment crops from high-investment crops with larger fertilizer jackpots.